

System Architecture & Technical Specification Document (US Market Standard)

1. Executive Summary & Purpose

The **Condominium Owners Association (COA) Management System** is an enterprise-grade SaaS platform tailored specifically for high-rise residential towers, luxury condominiums, multi-family housing complexes, and co-op communities operating in the United States.

Unlike single-family HOA communities or landlord-tenant rental properties, US Condominium Associations require specialized operational workflows, high-rise security protocols, reserve fund accounting compliance (e.g., Florida Chapter 718, California Davis-Stirling Act), and strict role-based access control.

This document outlines the system requirements, user role hierarchy, functional modules, and database multi-tenancy architecture designed to deliver a seamless experience for property managers, board directors, residents, and front-desk personnel.

2. Platform Portals & Multi-Tenancy Architecture

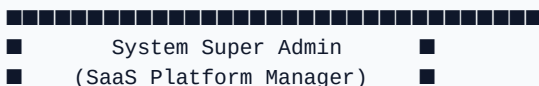
The platform provides dedicated, isolated portal entry points via the **Unified Portal Gateway**:

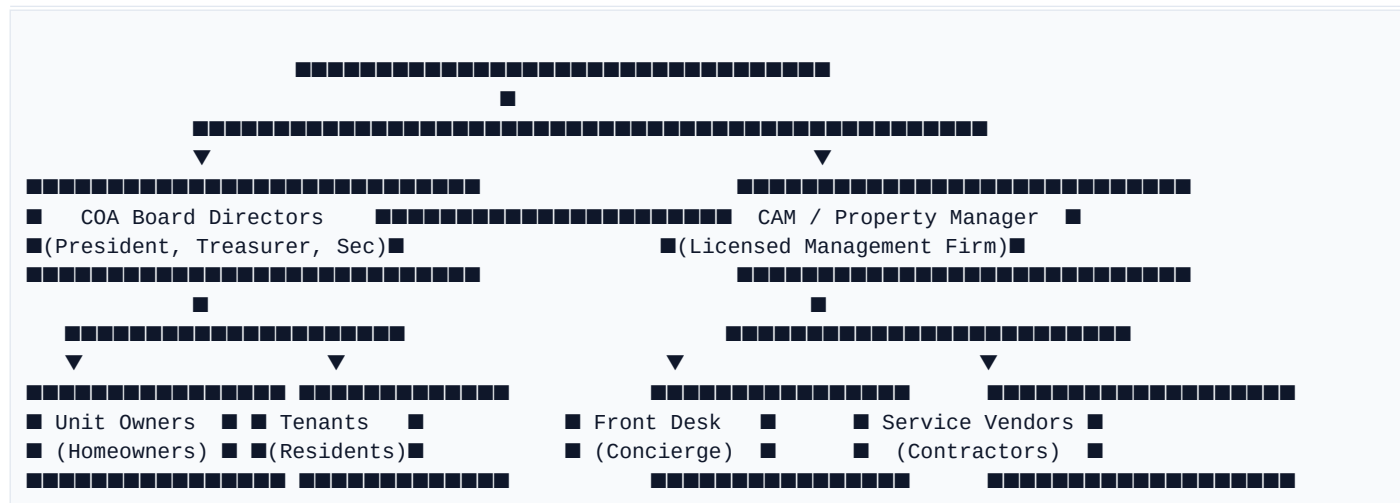
- **HOA Community Portal:** Designed for single-family subdivisions and horizontal gated societies.
- **Rental Property Hub:** Designed for landlords, real estate investors, and tenants managing lease terms.
- **Condo Management Portal (COA):** Specialized workspace for high-rise building managers, condo boards, unit owners, and front-desk concierge teams.

Data Isolation & Multi-Tenancy Strategy

- **Strict Tenant Isolation:** All database tables enforce a `community_id` foreign key. Queries are dynamically scoped at the ORM layer to prevent cross-association data leakage.
- **Role-Based Access Control (RBAC):** Fine-grained permissions ensure users access only the features allowed by their assigned role within their registered condominium association.

3. US COA User Role Hierarchy & Permissions





1. System Super Admin ([super_admin](#))

- **Scope:** Platform-wide administration.
- **Permissions:** System configuration, subscription billing, multi-community onboarding, global security monitoring.

2. Community Association Manager / Property Manager ([cam_property_manager](#))

- **Scope:** Licensed CAM or management company handling day-to-day COA operations.
- **Permissions:** Managing maintenance work orders, issuing Estoppel Certificates, vendor dispatch, COA dues invoicing, reserve fund oversight, amenity scheduling.

3. Board of Directors ([board_director](#))

- **Officers:** President, Treasurer, Secretary.
- **Permissions:** Approving annual operating budgets, special assessments, reviewing audit logs, executing vendor contracts, managing board meeting agendas/minutes (Robert's Rules of Order), reviewing ARC requests.

4. Condo Unit Owner ([unit_owner](#))

- **Scope:** Individual title deed holder of a condo unit.
- **Permissions:** Paying monthly COA assessment dues, casting proxy votes in board elections, submitting ARC remodel requests, authorizing tenant access, requesting resale disclosure packages.

5. Resident / Tenant ([resident_tenant](#))

- **Scope:** Occupant residing in a unit (owner-occupant or renter).
- **Permissions:** Generating visitor OTP gate passes, booking freight elevators for move-in/move-out, tracking front-desk deliveries, booking common amenities (pool, gym, party room), filing common area repair tickets.

6. Front Desk Concierge / Security Guard ([front_desk_concierge](#))

- **Scope:** High-rise lobby security and front-desk personnel.
- **Permissions:** Verifying guest OTP passes, logging incoming parcels/deliveries, confirming freight elevator move-in bookings, issuing guest parking permits.

7. Service Vendor / Contractor ([service_vendor](#))

- **Scope:** Third-party contractors (elevator engineers, HVAC specialists, plumbers, janitorial).
 - **Permissions:** Receiving work orders, updating repair job status, submitting invoices, uploading Certificate of Insurance (COI) documents.
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4. Key Functional Modules

Module 1: High-Rise Access Control & Guest Management

- **Digital Visitor Passes:** Residents generate QR codes or 6-digit OTPs for guests, food delivery, and service contractors.
- **Concierge Verification Desk:** Real-time scanner interface for lobby guards to validate visitor credentials and log guest entry/exit timestamps.
- **Guest Parking Permits:** Digital permit issuance mapped to specific unit visitor quotas.

Module 2: Freight Elevator & Move-In / Move-Out Scheduler

- **Move Slot Scheduling:** Calendar interface preventing simultaneous move-ins that disrupt elevator flow.
- **Move Deposit Management:** Automated collection and refund of refundable damage deposits for move-in/move-out events.
- **Pre/Post Move Inspection Logs:** Digital checklists for property managers to record hallway and elevator cab condition.

Module 3: Front-Desk Parcel & Delivery Tracking

- **Barcode Delivery Scanner:** Front desk staff scan incoming packages (FedEx, UPS, Amazon) to instantly match them with unit residents.
- **Automated Resident Alerts:** Push notifications and SMS alerts sent to residents upon package receipt.
- **Digital Sign-Out Ledger:** Touchscreen or QR sign-out verification when residents collect packages.

Module 4: COA Financial Ledger & Assessment Dues

- **Assessment Billing Engine:** Automated recurring billing for monthly COA dues via ACH (Stripe/Plaid integration) and Credit Card.
- **Reserve Fund Tracking:** Segregated accounting for General Operating Account vs. Capital Reserve Funds (roof, elevator, painting).

- **Automated Late Fees & Delinquency Notices:** State-compliant grace periods and automated late fee assessments.

Module 5: Estoppel Certificates & Resale Disclosures

- **Title Company Requests:** Automated portal for title companies and buyers to request official COA Estoppel Certificates.
- **Fee Collection:** Instant processing of expedited or standard Estoppel certificate issuance fees.
- **Dues Verification & Lien Checks:** Auto-populates outstanding balances, pending special assessments, and violation status.

Module 6: Architectural Review Committee (ARC) Approvals

- **Remodel Applications:** Unit owners submit interior structural/flooring/plumbing modification requests.
- **Document Attachment:** Upload architectural blueprints, contractor licenses, and liability insurance certificates.
- **Board Review Workflow:** Multi-stage approval by the ARC committee and Board of Directors.

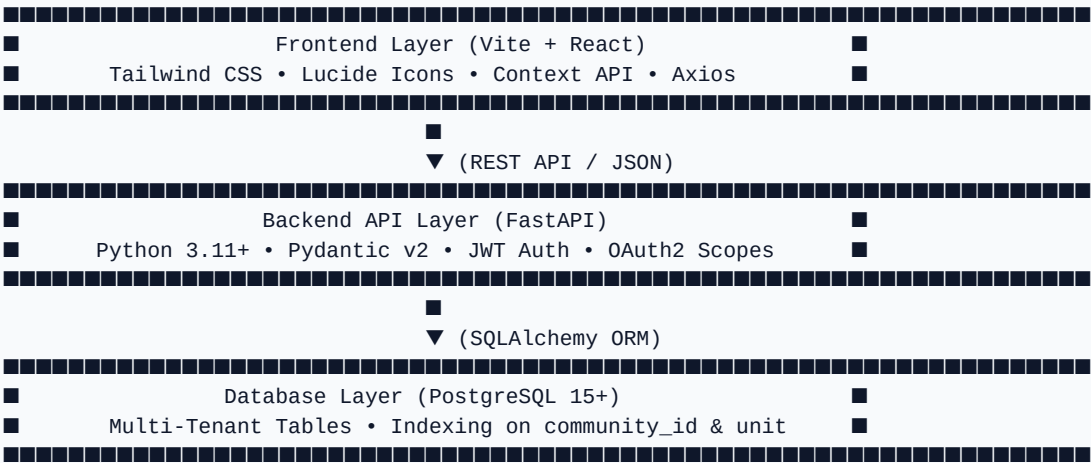
Module 7: Parking Spot & Storage Locker Allocation Manager

- **Asset Mapping:** Units mapped to designated indoor parking bays, EV charging stations, and storage lockers.
- **Dispute & Violation Logging:** Real-time logging of unauthorized parking or blocked access lanes.

Module 8: Bylaws AI Copilot & E-Voting

- **24/7 Rule Copilot:** Natural Language AI answering resident questions regarding building policies (e.g., pet weight limits, quiet hours, balcony rules).
- **Secure SHA-256 E-Voting:** Tamper-proof electronic voting for annual board elections and CC&R; amendment ratifications.

5. Technical Stack & Architecture Standards



6. Implementation Roadmap

1. **Phase 1:** Gateway integration

([PortalSelect.jsx](file:///d:/Vhoa_Management/frontend/hoa-portal/src/pages/auth/PortalSelect.jsx)) linking Card 3 directly to `/login?type=condo`.

2. **Phase 2:** Database Schema extension (`community_type = 'CONDO'`) and US COA role seeding (`cam_property_manager`, `board_director`, `front_desk_concierge`, `unit_owner`).

3. **Phase 3:** Core High-Rise Modules implementation (Visitor OTP, Freight Elevator Scheduler, Parcel Logger).

4. **Phase 4:** COA Financial Engine (Reserve Funds, ACH Payments, Estoppel Certificate Generator).

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Target Market: United States Condominium Associations & COA Management Companies

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